

BOUDROUSS RÉDA | CV

Software Engineer – OCaml, TypeScript & WebAssembly – Paris

github.com/rboudrouss
linkedin.com/in/rboudrouss
contact@rboud.com
rboud.com

Languages: French (Native), English (Professional), Arabic (Conversational)

+33 (0)7 55 78 85 16

Profile

Software engineer completing an MSc in Computer Science (Software Science & Technology) at Sorbonne University, focused on **programming languages**, **static typing** and runtime implementation. I work in **OCaml**, **TypeScript** and Rust. I brought a large OCaml/C/C++ codebase, the MOPSA static analyzer, to the **browser** via WebAssembly, and built a **local-first**, offline-capable mobile web app backed by a low-bandwidth-optimized server. I put a lot of weight on robustness and testing.

Projects

Mopsa-wasm - github.com/rboudrouss/mopsa-emcc

2026

- Brought a large **OCaml** application to the **browser** by compiling MOPSA, a static analyzer written in OCaml/C/C++, to **WebAssembly** with Emscripten, statically linking the OCaml runtime and all native dependencies (Clang frontend, Apron, GMP, MPFR).
- Worked across the **OCaml/JavaScript boundary**, running the non-reentrant runtime in Web Workers, implementing blocking stdin over SharedArrayBuffer + Atomics for an interactive REPL, and wiring virtual-filesystem preloading to an xterm.js terminal UI.
- Shipped a fully client-side [playground](#) (zero server-side compute) and a technical [write-up](#).

Reactant analyzer - github.com/rboudrouss/reactant-analyzer

2026

- Designed and implemented a static analyzer in **Rust**, based on abstract interpretation, for **React** web applications.

Experience

Quality Analyst Engineer (apprenticeship) - Hexaglobe (video streaming)

2025 – 2026

- Designed complete **test strategies** for zero-coverage legacy products, reaching **~3,000 unit / integration / E2E (Playwright) test cases** across 4 stacks (TypeScript, Python, PHP, JS).
- Built disciplined **AI-agent systems** (Claude Code) with sandboxed planner/implementer/reviewer orchestration, versioned guardrails, mandatory green tests and human review, producing **~720 integration tests** in 3 days.
- Built **GitLab CI/CD** pipelines and shared CI components consumed by 4+ repositories, plus objective video-quality oracles (VMAF, freeze/black-frame detection) to validate a live encoder.

Technical Lead, then President (2024–25) - Association Play Sorbonne Université

2023 – 2026

- Built a **local-first**, offline-capable mobile web app with secure authentication, a backend optimized for **low-bandwidth** clients and a database tuned for high-traffic spikes.
- Designed and operated a virtualized **infrastructure** (Docker servers, CI/CD, networking). As president, led **100+ volunteers** for a cultural event welcoming 10,000+ visitors.

Education

MSc in Computer Science (Software Science & Technology) - Sorbonne University

2024 – 2026

- Coursework: **static typing**, design & implementation of programming languages and runtimes, **concurrent & distributed programming**, static analysis by abstract interpretation.

BSc in Computer Science - Sorbonne University

2020 – 2024

- Algorithms, computer architecture, operating systems, networks, databases, with team projects in C, Java, OCaml, Python and TypeScript.

Skills

Languages: OCaml, TypeScript, Rust, Python, C, SQL

Topics: WebAssembly, local-first & offline architectures, testing (Playwright, Jest), GitLab CI/CD, Docker